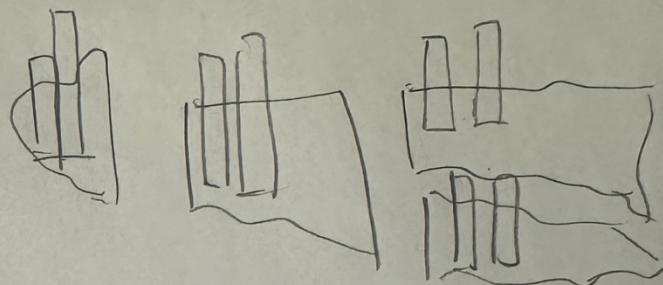


IDE A:

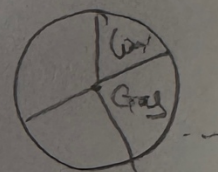
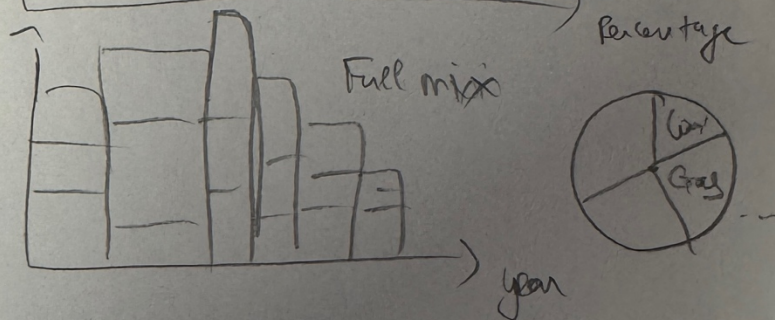
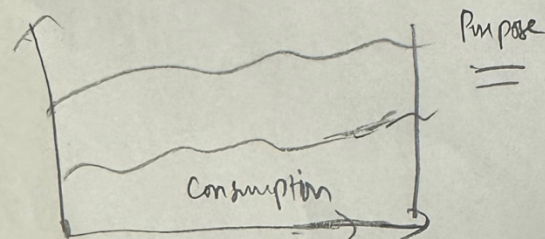
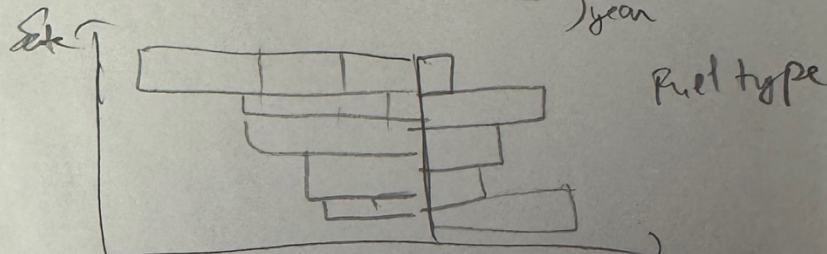
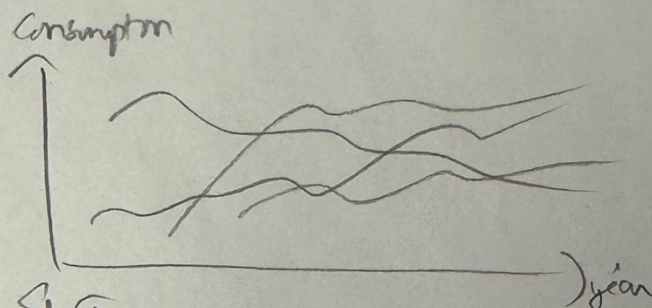
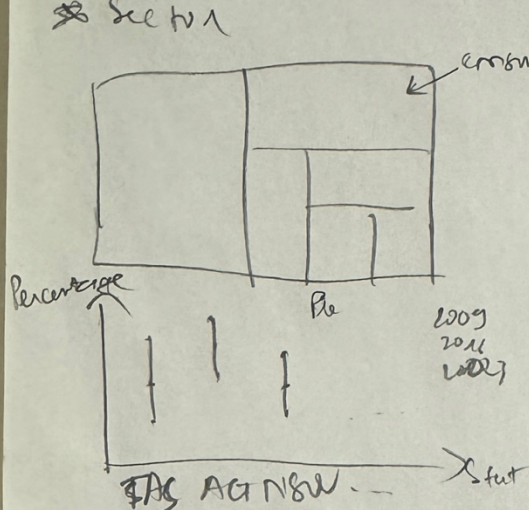


Major Power Station

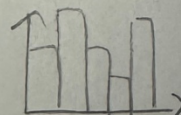
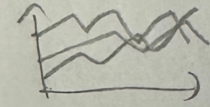
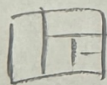
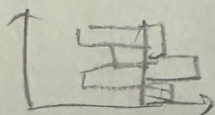
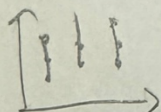
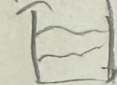


Consumption by state

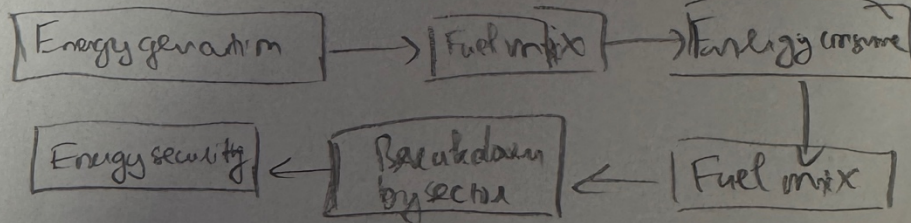
Section 1



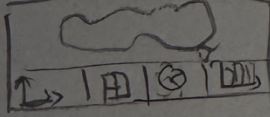
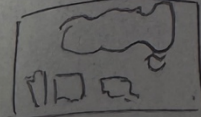
Filter



CATEGORY



COMBINE AND RE FINE



QUESTION

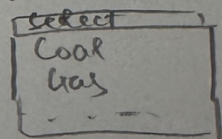
- 1, Can I create a separate map for each state
- 2, Does the data contain the required attribute.

LAYOUT

Title: Energy consumption
in each state
Author: Gia Minh Nguyen
Date: 2/10/2024
Sheet: FDS 2
Table: FIT 3179 Data Visualization

OPERATION

- Might have a zoom slider in the Australian map
- Highlight bar when a specific fuel is chosen or have a dropdown filter



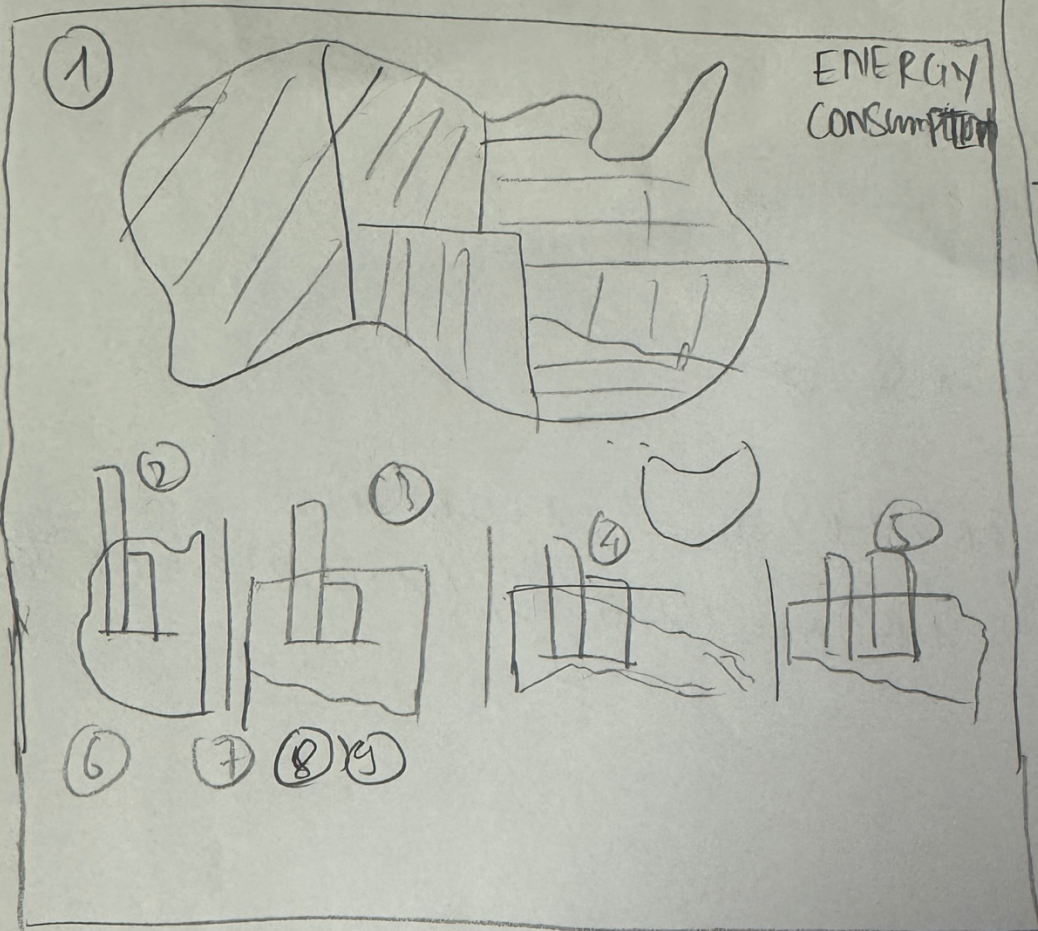
DISCUSSION

- Pos:
- Have clear look at each state
 - Have general info about the energy consumption

Cons:

- The state map be hard to visualize
- Might take a long time to progress

ENERGY CONSUMPTION



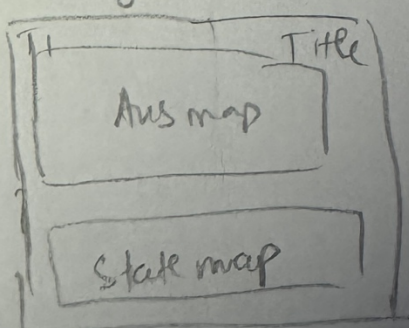
① Australia choropleth map show the overall energy consumption in Australia

② → ⑨ Bar chart with state map in back ground show the fuel mix of energy consumption

FOCUS

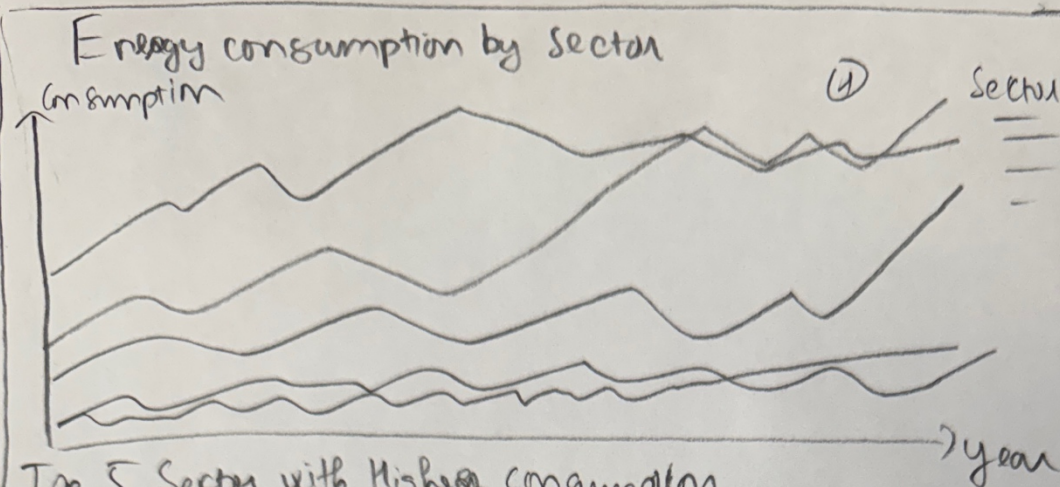
Main focus in the Australia map, and State map will provide more detailed information about the fuel mix of consumption

Arrangement

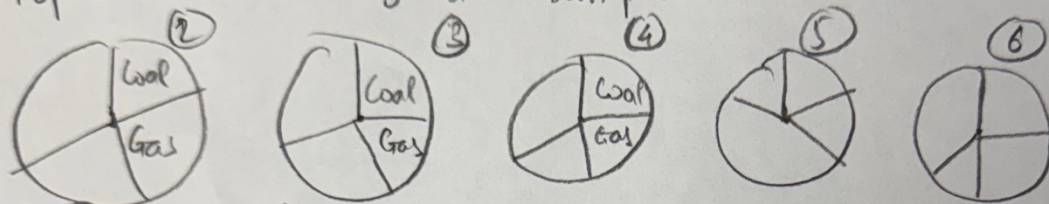


LAYOUT

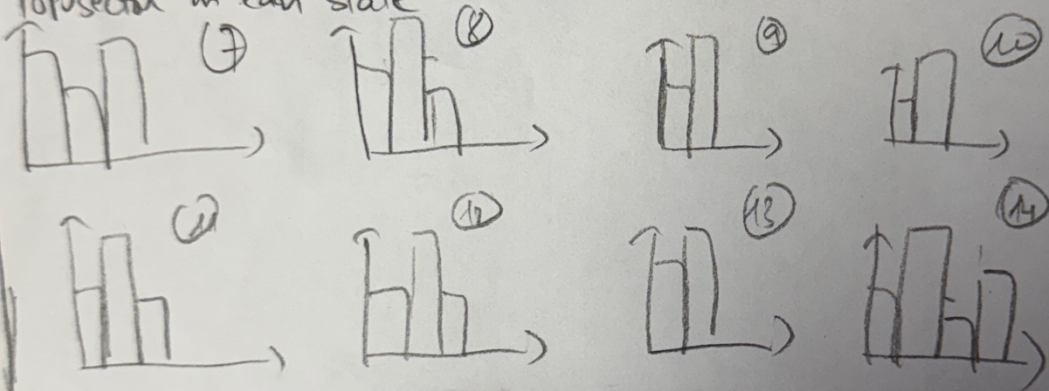
Title: Energy consumption by sector
 Author: Gia Minh Nguyen
 Date: 02/10/2024
 Sheet: FDS3
 Task: FIT3173 Data Viz 2



Top 5 Sector with Highest consumption



Top 5 sector in each state



① The consumption by sector from 1985 - 2023, named 10 sector

② - ⑥ The fuel mix of the top 5 consumption sector

⑦ - ⑭ The energy consumption of the top 5 sector in each state

FOCUS

- Focus on top 5 sector with the highest consumption
- The first line chart will give general information about all sector

Operation

Filter by sector in the line chart

E.g.

Select
Minns
...

Filter by state to choose the bar chart

E.g.

Select
NSW
VIC

DISCUSSION

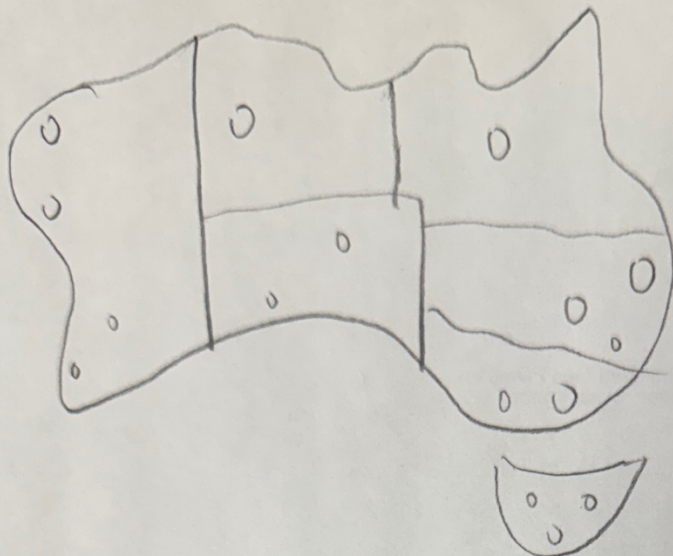
Pro:
 Have a clear look at the sector in Australia

Con:
 - might lack complex chart

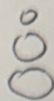
LAYOUT

ENERGY GENERATION

①



Energy Capacity



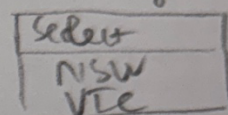
Energy Type
Renewable
Non-renewable

Title : Energy generation and fuel mix in Australia
Author : Gia Minh Nguyen
Date : 03/10/2024
Sheet : FDS4
Table : Delta Visualisation 2

OPERATION

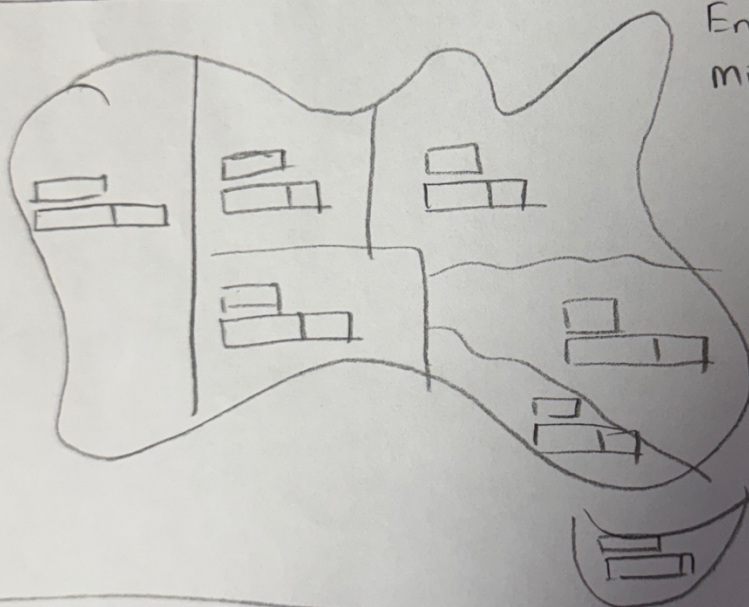
Full map
- Zoom in and filter
Zoom by slider

Filter by state, chosen
State will get in the center



Second map
- Highlight the
Selected energy type
When clicked in
the bar

②



Energy
Mix by State

DISCUSSION

Pro:

- Clearly show the
power plant location
and the fuel mix by
State

Con:

- Might lack the diversity
in the idioms.

- Might be hard to visualize
the second map

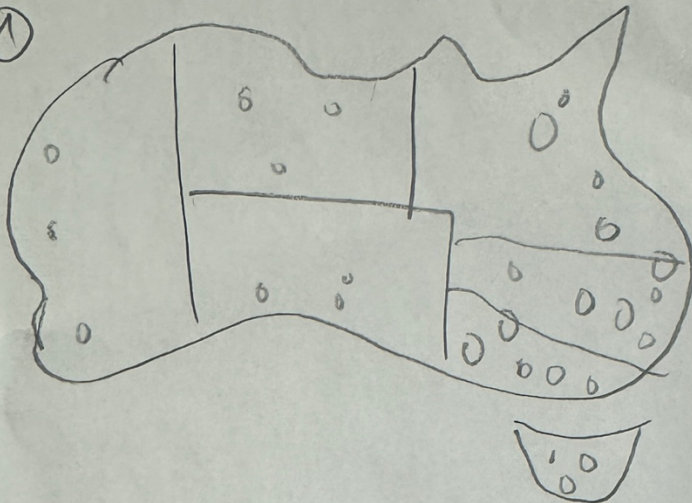
① This map show the location of powerplants in Australia, with size represent the capacity and color represent the energy type

② The map show the energy generation mix by state. The lower bar show the proportion of renewable vs non-renewable. The higher bar show the percentage of the most popular fuel type in that state

LAYOUT

Energy generation and consumption in Australia

①



Energy Type
Renewable
Non-renewable
Energy Capacity

Electricity Generation
Text

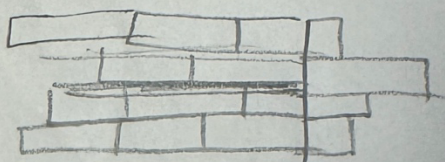
Percentage

②

Growth of Renewable
Electricity
Text

Energy Consumption
State

State



Fuel Type ③
Coal
Gas

Text

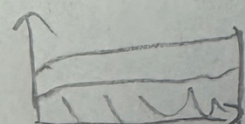
Percentage

Consumption

④ Sector
Mining

Text

Energy Security



Text

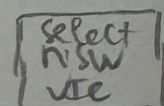
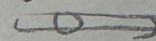
FOCUS

- No main focus, all charts are important
- Add annotation to chart to make it clearer
- ①, ②, ④ Use color to distinguish

Title: Energy generation and consumption in Australia
Author: Gao Minh Nguyen
Date: 03/10/2024
Sheet: FDS5
Task: FIT 3173 Data Viz 2

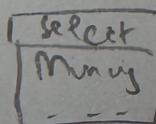
OPERATION

Zoom in and filter
by state



Highlight fuel type
Selected from legend

Filter by sector



Highlight the selected
area

DETAILS

- * Dependencies: Visual Studio code, R, Excel, Github, Vega lite
- * Estimated time and effort
 - 1-2 day to clean data
 - 4-5 days to build all chart
 - 1 day to insert text
 - 1 day to finalize everything
- * Specific requirement
 - Simplify data to avoid lagging or interesting